

Understandings of urban resilience meanings and principles across Europe

Lorenzo Chelleri^a, Aliaksandra Baravikova^{b,*}

^a School of Architecture, Universitat Internacional de Catalunya UIC, Immaculada, 22, 08011 Barcelona, Spain

^b Department of the Social Sciences, Gran Sasso Science Institute, Viale F. Crispi, 7, 67100 L'Aquila, Italy

ARTICLE INFO

Keywords:

Urban resilience
Resilience theory
Adaptation
Transformation
Sustainability
Resilient cities
Europe

ABSTRACT

In the last decade many papers have explored the challenges of defining and framing urban resilience. Academic literature has recently emphasised how urban resilience differs from urban sustainability, while urban policy agendas use these terms almost interchangeably. This paper explores European academics' and city practitioners' perspectives around urban resilience meanings and principles. It aims to enhance the understanding of the concept's evolution and to discuss the gap between theory and implementation. To do so, we conducted a survey among around 100 junior and senior European scholars from different fields and interviewed 24 city practitioners.

Both academics' and practitioners' perspectives have aligned with the latest conceptualisations of urban resilience. The respondents endorsed transformative and "bouncing forward" approaches, emphasizing that nowadays challenges stand within reconciling and combining transformation and resistance to change resilience perspectives. The more engineering, robustness and safety driven ones seem to be left behind. However, inconsistencies emerged when addressing implementation and resilience characteristics: interviews' responses showed how practices are still driven through robustness and safety-driven measures. The paper critically discusses these emerging conceptual misalignment and gaps to overcome in future research and practice.

1. Introduction

The concept of resilience is increasingly used as an organising principle framing scientific research, political discourses and entering cities agendas addressing urban vulnerability and sustainability. Throughout important international urban policy frameworks (as the UN New Urban Agenda, the Sustainable Development Goals, the Sendai Framework for Disaster Risk Reduction, among others) and programs supporting their implementation, urban resilience and sustainability have been used almost interchangeably (Elmqvist et al., 2019). Meanwhile, the conceptual development of urban resilience has followed different trajectories in different contexts, from being rooted in the field of climate change adaptation in Europe to disaster risk reduction in the US and similarly for Asian countries (Coaffee, 2013). This un-critical over-simplification in linking urban sustainability and resilience has contributed to many conceptual misalignments, inconsistencies and challenges for urban resilience implementation (Coaffee et al., 2018).

The latest advances in urban resilience research identifies a number of potential tensions and trade-offs, that might arise from the co-existence of policies seeking to develop robustness and transformations approaches (Chelleri, Schuetze, and Salvati, 2015; Chelleri, Waters, et al., 2015, Matyas & Pelling, 2015) or from political contestations

over who is made more resilient to what and at whose expense (Anguelovski et al., 2016; Vale, 2013). In their review of 25 published definitions of urban resilience, Meerow and colleagues observed that only a few definitions have explicitly addressed any of the above-mentioned issues (Meerow, Newell, & Stults, 2016). Since then there has been little progress regarding this issue. Three years later, Borie and colleagues argued the conflicts over resilience perspectives and values were not resolved but merely suppressed and hidden, warning that "where diversity is not acknowledged, debates over resilience will undermine the potential for more integrated policy and democratic decision-making" (Borie, Pelling, Ziervogel, Hyams, & Town, 2019, p. 204).

Such a malleability of the concept is perceived as beneficial by some, and problematic by others. From a theoretical point of view, resilience could be seen as a boundary object which fosters collaboration between disciplines (Brand & Jax, 2007). As Walsh-Dilley & Wolford (2015, p. 173) put, "this lack of ontological coherence unexpectedly makes resilience more productive as an object of inquiry than it would be if it were reduced to a standardised analytical framework or technical object." At the same time, other scholars condemned urban resilience as a buzzword, attributing its ontological ambiguity to the difficulty of translating the concept from theory into practice (Olsson, Jerneck, Thoren, Persson, & O'Byrne, 2015). The

* Corresponding author.

E-mail addresses: lcHELLeri@uic.es (L. Chelleri), aliaksandra.baravikova@gssi.it (A. Baravikova).

<https://doi.org/10.1016/j.cities.2020.102985>

Received 28 August 2019; Received in revised form 17 June 2020; Accepted 24 September 2020

Available online 27 October 2020

0264-2751/ © 2020 Elsevier Ltd. All rights reserved.

conceptual confusion between resilience and other similar concepts (such as sustainability) reduces the strength and consistency of its application. (Anderies, Folke, Walker, & Ostrom, 2013) Furthermore confusion surrounding the definition has been shown to create exclusionary policies, nullifying alternatives and hindering transformational change (Brown, 2014; Joseph, 2013). In light of these framing and implementation challenges, the paper explores critical gaps between urban resilience research and practices through the close examination of urban resilience meanings, principles and relations to similar concepts (particularly sustainability).

The overarching research question guiding this study is the following: “How is urban resilience conceptualised by researchers and practitioners across Europe?” Based on the interviews and survey results, the paper: i) investigates conceptual discrepancies around urban resilience in these two communities, ii) provides an in-depth focus on the relation between urban resilience and sustainability, and finally iii) discusses the implications regarding the current state of resilience implementation.

We explored the European context in order to get an in-depth focus and understanding of a specific region, with the intent to compare our results to similar studies from other parts of the world with different socio-economic and cultural backgrounds. The following sections will introduce: i) theoretical foundations of the above-mentioned tensions which underpin the survey and research design, ii) research design and methods and iii) critical exploration of the gaps between the academics' and practitioners' understanding of resilience, alongside with comparisons of European perspectives with a greater international context.

2. Theoretical background: embedded urban resilience tensions

2.1. Urban resilience: aiming at conservative or transformative approaches?

Academic literature has widely explored two main facets of resilience, metaphorically referring to either “bouncing back” or “bouncing forward” approaches. This distinction derives from disciplinary traditions of engineering and ecological (and socio-ecological) fields respectively (Holling, 1996). The Engineering perspective invokes bouncing back to a previous stable status after a shock prioritizing robust, absorbent and resistant capacities (Alexander, 2013). The ‘bouncing forward’ paradigm relies on innovation, change and transformation (Satterthwaite & Dodman, 2013) to mitigate future shocks and stresses.

Oversimplification of these opposite but coexisting perspectives on resilience has been explored in-depth by academic literature. The shift toward the bouncing forward paradigm (Manyena, O'Brien, O'Keefe, & Rose, 2011; Wenger, 2017), or transformative approach (Matyas & Pelling, 2015) has contributed to the integration of sustainability into disaster resilience, including methodological (Achour, 2015; Rodriguez-Nikl, 2015), community (Cutter, 2013) and specifically urban (Asprone & Manfredi, 2015) perspectives. This conceptual evolution intertwined with policy discourses labelling those shifting perspectives as a need to “build back better” (Mannakkara & Wilkinson, 2013). Without entering here the discussion about what “better” means (Khasalamwa, 2009), a similar conceptual evolution of resilience toward transformative approaches has happened in climate resilience (Pelling, 2011). Recent literature has investigated the linkages between climate adaptation and mitigation (Landauer, Juhola, & Söderholm, 2015), pointing out potential synergies and conflicts among these two different perspectives (McEvoy, Lindley, & Handley, 2006; Thornton & Comberti, 2013) and exploring the ways of linking them in climate planning practice (Grafakos, Trigg, Landauer, Chelleri, & Dhakal, 2019).

The relationship between transformative and conservative perspectives has been recently explored by resilience thinking theories through incremental or disruptive innovations in population development models (Pearson & Pearson, 2012). In development studies, the bouncing back or forward approaches have been translated as respectively “functional resilience” (aiming to manage change deliberately

and defining desired outcomes) and “dynamic resilience” (emphasizing uncertainty, evolution and non-linearity) (Tanner, Bahadur, & Moench, 2017). The latter has been also referred to as “evolutionary resilience” by planners (Davoudi, Brooks, & Mehmood, 2013) and regional economic scholars (Boschma, 2014; Simmie & Martin, 2010). The very fathers of resilience thinking (Walker & Salt, 2012) shifted their theories about ecological and socio-ecological resilience (Folke, 2006) increasingly and explicitly integrating transformational properties into their frameworks (Folke et al., 2010).

All of these theories and disciplines addressing resilience from different perspectives have informed, in neither structured nor integrated ways, our conception of urban resilience (Chelleri & Olazabal, 2012). Resilience implementation in cities usually followed a more engineered and simplistic perspective framed around security and protection (or resistance) to change, guarantee business continuity and socio-economic status quo (Tanner et al., 2017). This approach has been more appealing to policymakers because it does not threaten power structures and regimes (Smith & Stirling, 2010; Vale, 2013). Transformation has been integrated in most of the current definitions of urban resilience from academia (Meerow et al., 2016) and through policy implementation frameworks (The Rockefeller Foundation, 2014, CRPP UN Habitat). Nevertheless, the call to frame urban resilience thinking around transformative change (Elmqvist, 2014) is extremely problematic to put into practice. Only a few studies have explored the synergies, conflicts and trade-offs between conservative, adaptive and transformative approaches to resilience that emerge in their integration and implementation (Chelleri, Schuetze, and Salvati, 2015; Chelleri, Waters, et al., 2015; Matyas & Pelling, 2015; Tanner et al., 2017; Coaffee et al., 2018). In some cases, it is easy to determine which city characteristics need to become more robust and better protected, and which characteristics need to be transformed altogether. But often the co-existence of adaptation and transformation potential results in complex political and management choices, implying critical spatial and temporal trade-offs (Chelleri, Schuetze, and Salvati, 2015; Chelleri, Waters, et al., 2015). Since the politics of resilience emerges as one of the key factors in determining the path of urban resilience implementation (Vale, 2013), it is necessary to better understand how to enrich the dichotomy between conservative or transformational approaches in practice. This necessitates asking questions about which kind of transformation (or adaptation) we want (Meerow & Newell, 2016), who is the “we” in community resilience (Mulligan et al., 2016) and how this relates to urban sustainability goals.

2.2. Urban resilience and sustainability: how these concepts and their principles align or diverge?

When introducing the limitations of resilience as a useful concept driving urban transformation toward sustainability (Gillard, 2016), it is worth deepening the understanding of urban resilience thinking to move toward its practical implementation. This requires reframing its initial conceptual normative and positive connotations that are associated with its metaphorical meaning of “being adaptable”. After some decades of inspiring and positive resilience thinking (Walker & Salt, 2006), related to urban disasters (Vale & Campanella, 2005) and climate change (Newman, Beatley, & Boyer, 2009), a criticism from academia started to emerge (Friend & Moench, 2013; Pizzo, 2015; Satterthwaite, 2013; Weichselgartner & Kelman, 2015). It revealed how urban resilience implementation can sometimes be environmentally unsustainable (i.e. desalination plants as adaptation to droughts and the consumption of fossil fuels and socially unjust (i.e. green-induced gentrification) (Anguelovski et al., 2016). Since the early criticism of urban resilience concept is not always driving toward positive (or desirable) transformative outcomes (Derissen, Quaas, & Baumgartner, 2011; Chelleri & Olazabal, 2012), there has been an increased exploration of: i.) the relationships between resilience and (environmental) sustainability (Redman, 2014; Zhang & Li, 2018) and ii.)

linkages between resilience and social justice (Doorn, Gardoni, & Murphy, 2018; Eakin et al., 2017; Matin, Forrester, & Ensor, 2018; Rodina et al., 2017).

Both these research lines were underpinned by the realisation that neither adaptation nor transformation lead to sustainability per se, despite their metaphorical meaning of bouncing forward (Elmqvist et al., 2019; Wolfram & Frantzeskaki, 2016). This poses an important task of better aligning theories, policy frameworks and practices. Academic communities agree on conceiving of urban resilience as not always positive or as a desirable concept, which implies trade-offs (Chelleri, Schuetze, and Salvati, 2015; Chelleri, Waters, et al., 2015, Tanner et al., 2017) and thus differs from sustainability (Elmqvist, 2017). However, multilateral agencies and policy frameworks interpret resilience as overlapping with, and almost substituting sustainability (Weichselgartner & Kelman, 2015). A recent perspective paper published by Elmqvist and colleagues in *Nature Sustainability* points out how urban resilience and sustainability are both poorly understood, narrowly defined and interchangeably used even by UN Habitat New Urban Agenda and the UN Sustainability Development Goals which are supposed to be the global policy guidelines driving urban future (Elmqvist et al., 2019).

Such misinterpretation is due to different factors. Among these, sustainability per se has been simplistically implemented in cities, typically through the introduction of densification policies, the increased efficiency in resource consumption and technocratic measures (Checker, 2011; Neuman, 2005). Moreover, the unresolved disagreement about characteristics of urban resilience led to various trade-offs among its guiding principles, as illustrated in the following lines. This has indirect implications for how resilience relates to sustainability. The literature is rich in references proposing specific indicators and principles for enhancing the resilience of critical infrastructures (Lhomme, Serre, Diab, & Laganier, 2013), urban energy grids and supply (Sharifi & Yamagata, 2016), addressing flooding resilience through urban design (Liao, Le, & Van Nguyen, 2016), operationalising climate resilience (Leichenko, 2011), hazard and disaster mitigation (Godschalk, 2003) through the lens of communities (Cutter et al., 2008) or public investment, management and urban development (Jha, Miner, & Stanton-Geddes, 2013). Within these papers the “resilience of what to what” (Carpenter, Walker, Anderies, & Abel, 2001) is clearly defined. This is no longer the case, however, when we shift from specific resilience to generic resilience measurement frameworks. As Schipper and Langston (2015, p. 21) stated, “despite disagreements over how to define or delineate the concept, it is hoped that there will be some sort of general convergence on the characteristics of resilience” (Schipper & Langston, 2015, p. 21). This scholarly disagreement regarding the characteristics, principles and indicators for urban generic resilience has been further explored by Sharifi et al. (2017). He found that attributes and principles of different facets of urban resilience interact with each other, which results in a complex matrix of potential synergies and conflicts which cannot per se drive urban transformation. Resilience principles therefore do not always function in isolation (Sharifi, 2016), which would be the case for social-ecological resilience. For example, redundancy is a fundamental characteristic of resilience in social-ecological systems which contribute to the adaptive capacity and sustainability of a system (Berkes et al., 2003). It might decrease the competitiveness of a city in socio-economic terms but increase resilience of an infrastructure network or benefit urban ecosystem services provided by green infrastructure. Therefore, the paradox is that while a certain degree of redundancy always benefits the resilience of social-ecological systems, in cities it could imply positive, negative and neutral results that might decrease or increase adaptive capacities of each subsystem (Chelleri, 2016).

Another potential trade-off between resilience and sustainability is that resource efficiency may reduce diversity and redundancy, both of which are key features of resilience (Elmqvist, 2017). Dodman et al. (2017, p. 101) also emphasised possible tensions between social

resilience and resource efficiency emerging when “mandatory energy efficiency standards might increase the costs of consumer goods so that low-income households struggle to cover the upfront costs of, for example, air conditioners or light bulbs.” At the same time, however, they noted that “some short-term strategies to strengthen economic resilience may increase the pressure on ecosystems and thus lead to inefficient resource use.” Because of this lack of consensus on indicators and approaches, this paper sheds some light on these aspects by providing the current perceptions from scholars and practitioners across Europe.

3. Research design, methods and limitations

The research design was inspired by the Meerow and Stults (2016) study which explored practitioners' understanding of climate resilience and its principles across the US. Our paper responds to that research by discussing the perceptions of practitioners across Europe, complementing them by responses from academics. We designed a similar questionnaire which was sent to junior and senior scholars, experts in resilience coming from different fields, and the participants of the international conference Reframing Urban Resilience Implementation held in Barcelona (December 2018). The survey was conducted from April to June 2018 and we received 97 entries. At the second stage of our research we conducted 24 in-depth semi-structured interviews (questions followed the survey main sections) with key local practitioners working on city resilience (list in Appendix B) and covering quite a consistent part of Europe (Sweden, Finland, Denmark, the Netherlands, UK, Spain, Portugal, Italy, Greece, Belgium, Germany, Slovenia, Poland and Slovakia).

The literature review section of this paper assisted in framing a 12-item survey, divided into two parts. The first aimed to establish respondents' profiles and experiences in resilience, and the second part contained questions as to i) whether resilience is more associated with conservative or transformative approaches, ii) how to align these two different approaches; iii) what are the resilience characteristics and principles (to double check the consistency of responses about conservative or transformative approach), and iv) a set of closed questions about the relationship between resilience and sustainability (and other similar concepts such as smart cities). Closed-ended survey responses were quantitatively analysed using the Qualtrics platform. Open-ended responses were exported to NVivo software, where we performed a qualitative content analysis while taking into account the frequency of specific words used (Saldana, 2009). The full questionnaire and the details of the coding procedure are available in Appendix A.

Before looking into the results, we briefly explored the profile of the questionnaire respondents. As illustrated in Fig. 1A, 82% of the respondents came from academia, while the remaining part is a mix of public sector, industry, activists and NGOs. Overall, 11 different disciplinary fields were represented as could be seen in Fig. 1B. Urban planning was the most popular (21%) and engineering the less represented (3%). Most of the respondents stated they had at least five years of experience in resilience, while only 20% said their work dealt with resilience but not focused solely on it.

Providing regional overviews about concepts is subject to inevitable limitations, being the main one the representativeness of the sample. We tried to overcome this issue by balancing junior and senior scholars, covering different disciplines and countries of affiliation. Because of the relatively recent inclusion of the concept of resilience into EU policy and practices, we reasonably believed that our sample of 97 responses could provide an overview of definitions and perspectives. The same limitations and strategies used to overcome them apply to the practitioners' sample which consists of 24 different profiles across Europe. Furthermore, in the interviews with practitioners we aimed not to provide an overview of European resilient cities (which would have been a different research question and thus design) but to compare scholars' and practitioners' perspectives on urban resilience approaches

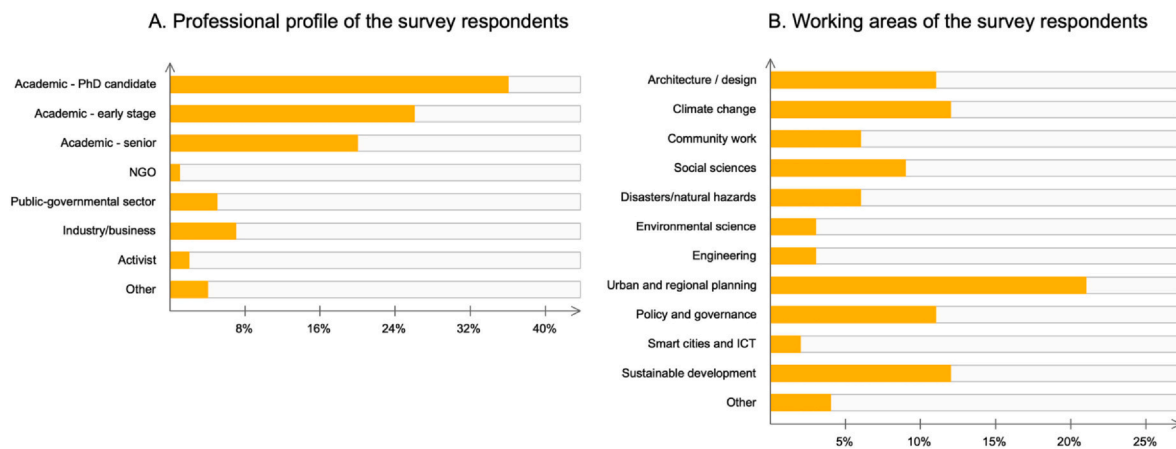


Fig. 1. Survey respondents' profiles.

and principles for its implementation. Last but not least, this research design provides only a snapshot of urban resilience perspectives. Perceptions are not static but change and evolve through time, likewise, resilience is a rapidly evolving topic. To note, the research by [Meerow and Stults \(2016\)](#) which acted as a point of comparison for this study, was conducted in 2015, and this one in 2018.

4. Results: how urban resilience is conceived across Europe?

In this section we present the results of the survey, comparing them with the practitioners' perspectives. They are organized in two subsections, the first focusing on resilience perspectives and the second on characteristics and principles.

4.1. Resilience and transformation: navigating through different although coexisting approaches

4.1.1. Bouncing back, forward or both?

One of the clear results from this research is about positioning the evolution of the resilience concept in reference to its “bouncing back or forward” approach. As illustrated in [Fig. 2A](#), the 72% of the survey respondents indicated that resilience is both about bouncing back and forward, while the 22% said it is only about bouncing forwards and only the 6% said it's about bouncing back.

This result confirms European scholars' perception is in line with the recent theory trends discussed in our literature review. They left behind the old bouncing back approach while embracing a more evolutionary perspective of resilience. Luckily, the practitioners' perspectives also confirmed this shift, as bouncing forward ranked higher than bouncing

back (see [Fig. 2B](#)). “We have the ambition to bounce forward [...] to be prepared to shocks but in the meantime also see how they can improve our city” (city practitioner, the Netherlands 2019). A popular line of reasoning was that crises provide opportunities to enhance resilience: “every disaster brings a possibility to transform the city and adapt to new circumstances, and develop in new direction” (city practitioner, Poland, 2019).

However, more than half of the interviewees (13 out of 24) stated resilience was not only about bouncing forward but about both bouncing back and forwards. “As changes are huge, bouncing back alone will probably not be the safer option. You need to bounce back, but while looking how to reduce future risks” (city practitioner, Denmark 2019). Finally, two interviewees said resilience is only about bouncing back. In the words of a practitioner from Italy, “since the term resilience comes from the materials science and engineering, its very original meaning would be about the ability to bounce back [...] it is in practice only the capacity of a city to cope with its risky situations.” Even though he admitted that “I noticed Rockefeller Foundation introduced the concept of ‘Resilience Dividend’ to promote bounce forward capacities, that could therefore fit into a wider definition of resilience: after all, resilience has become an umbrella concept”. Some initial contradictions emerged among the proponents of the bounce back paradigm: “resilience is about bouncing back because cities need to anticipate and transform themselves to be resilient for eventual shocks, otherwise they will not recover from disasters” (city practitioner, Spain 2019). This sense of inconsistency, while defining resilience, is probably induced by having a variety of frameworks, principles and approaches for implementation whereas a universally agreed upon framework or clear set of guiding principles may be lacking. ([Sharifi & Yamagata, 2016](#);

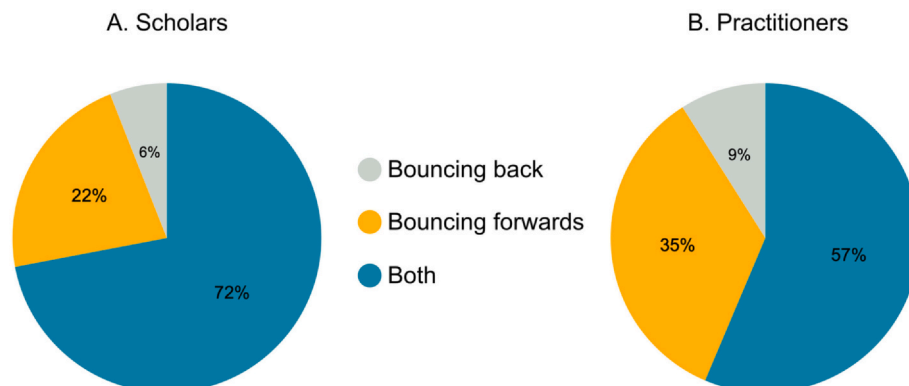


Fig. 2. Answers of scholars and practitioners to the question whether resilience is primarily about bouncing back, forwards or both. (Source: authors' elaboration)

Barriers to reconcile 'bouncing back' and 'forwards' perspectives (according to the survey participants)

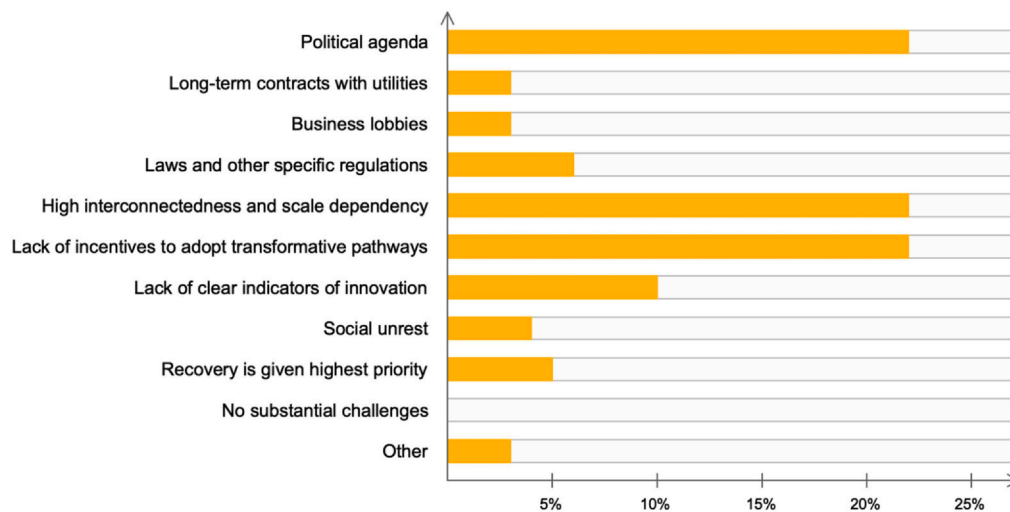


Fig. 3. The barriers to reconcile 'bouncing back' and 'forwards' (based on the survey).

Coaffee et al., 2018).

4.1.2. Concepts alignment in practice: how to implement something with potentially conflicting meanings coexisting within its definition?

This popularity of combining 'bouncing back' and 'bouncing forward' approaches calls for a careful consideration of how this complex concept would be implemented. According to survey respondents, this undertaking is not devoid of challenges. As illustrated in Fig. 3, the challenges and barriers to aligning bouncing back and forward paradigms were mostly related to political agendas being framed around immediate or short-term goals, looking to keep the status quo of funding toward infrastructure, services and businesses. Another prominent challenge was the 'high interconnectedness and scale dependency' which refers to the feeling of being too dependent on regional or national scales in terms of markets, contracts and also the regulatory power cities lack. Many qualitative survey responses reported that "it's difficult to innovate and scale-up innovation alone, from the local level." Also a "lack of financial and/or political incentives to adopt transformative pathways" ranked among the top barriers for integrating transformation and robustness (20%).

Other issues (see Fig. 3) included a higher priority toward recovery and the lack of clear resilience indicators related to innovation. The latter means not distinguishing between incremental innovation and conservation (or enhancing efficiency) and disruptive innovation indicators driving systemic change. Regulations and social unrest were also listed among the obstacles. Long-term contracts with utilities and business lobbies received only limited attention compared to more generic issues (both scoring only 2%), which could perhaps be explained by theoretical perceptions of scholars involved in the survey.

Unexpectedly, most of the interviewed practitioners did not see any substantial issues in implementing these divergent perspectives on resilience simultaneously. The few exceptions include a practitioner from Italy who mentioned the issue of "framing innovation" (lack of enhancing skills of individuals and groups for transformative changes upscaling), and a practitioner from Slovakia who described how numerous local constraints (i.e. land regulations and bureaucracy) made transformation difficult to meet resilience implementation. A more common position can be illustrated by the following statements: "if you plan for bouncing forward, then you can expect that you have a very quick response in bouncing back" (city practitioner from Greece, 2019) or "In order to be a transformative city, it means you take good things

and make them useful. Meaning that you are also being able to be prepared if something happens. So I don't see a big contradiction" (city practitioner from Sweden, 2018). Interestingly, one respondent linked these two perspectives through the idea of learning capacities: "I think you first need to have a bit experience, understanding what it means to bounce back, to be able then to plan how to advance" (city practitioner from Slovakia, 2019).

One of the key learnings emerging from the interviews was that for most practitioners 'bouncing forward' implied the capacity to learn and improve while recovering and enhancing potential responsive capacities. This is a valuable insight for reframing future research on urban resilience. However, in no case 'bouncing forward' has been associated with innovations for radical urban transformations, as was often described by academics and the literature.

4.1.3. Beyond bouncing back or forward, how these perspectives relate to urban sustainability?

The overall narrative in both surveys and interviews can be summarized as 'resilience and sustainability do not always share the same goals, but nevertheless they do not contradict each other.' Indeed, most of the survey respondents (74%) said that resilience does not always have the same goal as sustainability (Fig. 4A); more than a half (60%) believed resilience and sustainability do not contradict each other (Fig. 4B). Nevertheless, it is difficult to establish a clear understanding of the links between these concepts. Among those who don't see resilience and sustainability as conflicting, about half (51%) considered resilience as a prerequisite for achieving sustainability, only 15% disagree with that and 34% said that 'not always' (Fig. 4C).

Curiously, only 16% of the surveyed European academics agree that resilience is more appealing to policymakers than sustainability, which stands in contrast to the international literature (see, e.g. US perspective in Redman, 2014). Slightly more than a half of our survey respondents (58%) said "not always", which might suggest the increased confusion in the perceptions of resilience and sustainability.

Unfortunately, the interviews with practitioners did not clarify the linkages between these two terms. While about 60% of them considered that resilience and sustainability have aligned goals, the remaining 40% put forward examples explaining how the concepts' goals significantly differ. Still, some of those who think the concepts' goals are aligned saw sustainability as an umbrella concept in which resilience is nested. They talked about how these concepts "are very similar, but sustainability

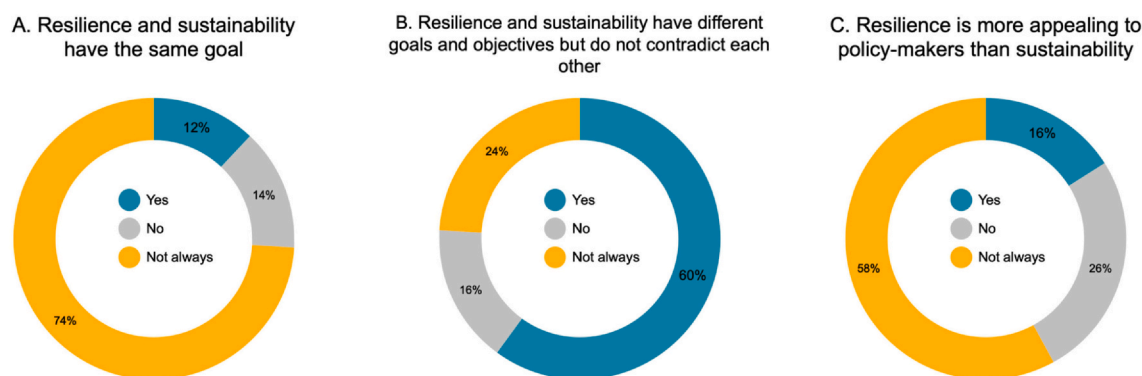


Fig. 4. Relations between resilience and sustainability (based on the survey).

Characteristics of a resilient city based on the survey (counts)

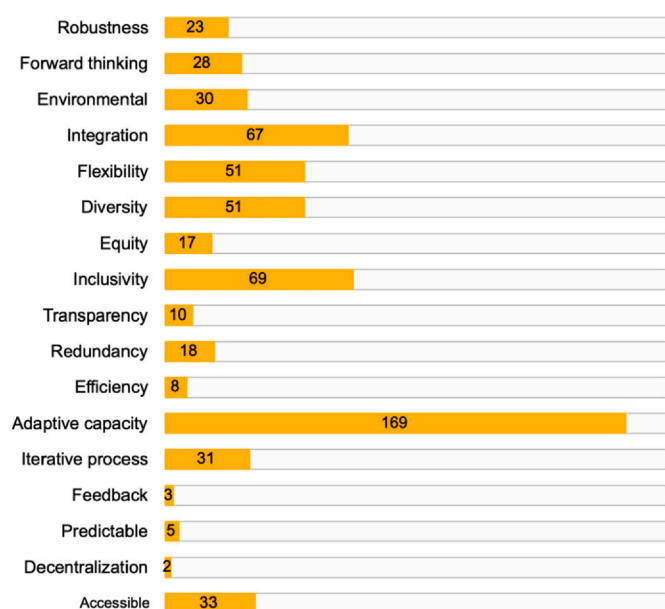


Fig. 5. Number of characteristics of a resilient city based on coding (among survey participants).

goes further [...] making sure future generations can live as well or better as we do" (city practitioner, the Netherlands, 2019), that "resilience is the basis of sustainability" (city practitioner, Slovakia, 2019), or argued that "there are no differences between the two concepts. We use sustainability to define the global long-term goals, and resilience to define our local specific short-term actions" (city practitioner, Italy, 2018).

Others saw *resilience as the broader term*. In the words of a practitioner from Poland, "resilience is the next step in city's development, following sustainability". Those who thought resilience and sustainability have different goals also suggested that "working in resilience is a consequence of not having worked enough on sustainability" (city practitioner, Spain, 2019), or described how unsustainable issues can be very resilient: "a sustainable place should be resilient, but you don't need to be sustainable to be resilient. An unsustainable solution like heavily engineered flood defenses can enable a community to be more resilient" (city practitioner, UK, 2019). According to the city practitioner from Italy (2019), "there are cases where a policy can be environmentally unsustainable yet that measure is definitely increasing resilience. As an example, if we want to safeguard some low-lying cities

from increasing sea level rise, we cannot limit the actions to just soft and environmentally friendly solutions, but also intervene with large engineered solutions. There are no other options in these cases, and as a result, resilience and sustainability are sometimes even in conflict." And more about maintaining the unsustainable status-quo while promoting resilience: "In Thessaloniki we are not sustainable in term of mobility, but we are building resilience for emergency cases if something happens. In this case we are building resilience for an unsustainable system to persist and maintain its functions" (city practitioner, Greece, 2018).

Interestingly enough, a number of interview respondents saw resilience as a more dynamic and socially-oriented concept: "Sustainability has some kind of physical, technical aspect to it, while resilience has much more social context to it" (city practitioner, the Netherlands, 2019). They also felt that resilience provides more clear guidance: "sustainable development can and should have concrete goals and actions in addition to the important more general goals, but often that is not the case (yet). Resilience, on the other hand, at least as adaptation planning, has nowadays concrete actions in some sectors in addition to the more general goals or strategies (city practitioner, Finland, 2018).

4.2. Urban resilience characteristics: which principles are driving implementation?

To test the consistency of the above-mentioned responses we looked into how the urban resilience principles and characteristics, brought up in the survey and interviews, relate to the generic definitions of resilience. When asked to list urban resilience characteristics, survey respondents provided various types of answers, ranging from a strict list of principles to sentences explaining them. The results of the coding and content analysis of their answers are summarized in Fig. 5. Overall, socially-oriented and soft characteristics (i.e., capacities) turned out to be more popular than technology and hard characteristics (i.e., referred to infrastructures and buildings).

The answer 'adaptive' won by a huge margin, perhaps due to its metaphorical meaning that avoids a specific definition of the characteristic. It was followed by "inclusivity," "integration," "flexibility" and similar characteristics which were mentioned more than "robustness." Results, therefore, reflect the idea that resilience measures and implementation should be based on open processes focusing on social justice. Similar answers were given to the open-ended questions about the strategies and characteristics of urban resilience. However, "flexibility" and "diversity" were also mentioned in relation to "enhancing across scales and sectors collaborations." For some, "diversity" meant "good social mix," while for others diversity was referred to land use and urban planning. Among the least popular characteristics turned was "decentralisation" (maybe because it is perceived as being too engineered, although in fact it is a trait of most ranked characteristics), "predictable" (seeing resilience as a command and control practice) and "feedback" (more in line with social-ecological literature than urban

studies one which usually refers to participation, inclusivity, process and learning, rather than feedbacks among subsystems).

In the interviews, most of the practitioners named characteristics that are in line with socially-centred perspectives on resilience (equity, empowerment, inclusivity); the second big group of responses referred to risk preparedness-related characteristics. Among the most popular concepts in the first group were “social justice”, “inclusive city”, “small income differences, equal opportunities”, “little segregation”, “participation in planning processes”, “cooperation among stakeholders.” A practitioner from Italy puts it: “without sufficient awareness and participation by citizens, no resilience policy is seen as acceptable. Awareness is the foundation of the resilient city”. Similarly, in the words of a practitioner from UK, “resilience is about the empowerment of communities to deal with the shocks and changing circumstances that hit them on a regular basis.” Reflecting on the role of communities, one of the city practitioners from the UK (2019) said it is necessary to “enforce and build resilient communities, risk aware and not risk adverse, and prepared to work in a collaborative way to help themselves.” In addition, “to have a resilient city you require empowered, knowledgeable citizens, trusting officials and key understandings on who is delivering what. Less resilient communities rely on ‘others’ to solve problems for them, feel that things happen to them and that they have limited control over actions” (city practitioner, UK, 2019).

The second largest group of responses was focused on the importance of disaster preparedness, featuring characteristics such as “risk governance,” “anticipatory capacity,” “well-prepared” or “understanding future risks.” Most of them are key resilient characteristics and capacities necessary to have a “functioning, well maintained infrastructure” (city practitioner, Finland, 2019), or “robust critical infrastructure” (city practitioner, Sweden 2018). According to a city practitioner from Slovakia, you need to have “a very good knowledge base of what is vulnerability, which areas, who is vulnerable, what is the risk and where is the risk. You have the data based on which you can move forward and do the planning”. Importantly, all the practitioners mentioned characteristics from both groups, what differed was the extent of elaboration each group received.

Comparing the results from the survey and practitioners' interview responses, both groups listed and built discourses to explain resiliency characteristics by using metaphorical, or generic concepts, rather than clear physical or management related characteristics (i.e. decentralised infrastructure networks, integration of nature in cities, more adaptive regulations, or co-management strategies for urban services, etc. In a way, we could say that when asked to address the core aspects of urban resilience, both groups illustrated the linkages to other concepts, for instance by listing the notions of inclusivity, equity and empowerment which can be associated with distributive, procedural and recognition aspects of social justice.

5. Discussion and conclusion

5.1. Gaps between theory and practice

Quite a confusing landscape of understanding emerged from the survey and interview results, especially with regard to practitioners' perspectives and implementation consistency between definitions and characteristics of urban resilience. We could start by comparing our results to the previous research conducted by Meerow and Stults (2016) about the US practitioners' perspectives on climate resilience. Before embarking on the comparison, it is worth mentioning that in Europe mobilisation of the urban resilience concept developed in the context of climate change adaptation (Coaffee, 2013) and was therefore linked to broader environmental and sustainability concerns. On the contrary, in the US (similarly to Asia) it emerged in relation to disaster risk reduction and security. This background is in line with the Meerow and Stults (2016) study indicating the preference for a ‘bouncing back’ approach with an emphasis on robustness as a key characteristic. Our results

suggest a significant difference between the EU and US perspectives, since both researchers and practitioners across the EU tended to define urban resilience as linked to bouncing forward or a concept integrating bouncing forward and back approaches.

Whereas both the US and EU practitioners mentioned ‘adaptive capacity’ among the most popular characteristics of urban resilience, the difference is that across the EU it was of primary importance, while in the US it falls behind robustness. However, it is not very clear yet how the popularity of ‘adaptive capacity’ as a main resilience characteristic is translated into practices toward ‘bouncing back’ or ‘bouncing forwards’ approaches. Indeed, the metaphorical strength of adaptive capacity – a vague and inconsistently defined notion in itself (Hinkel, 2011) – contributes to drive its implementation challenges, since it spans over different levels of decision-making (Eakin, Lemos, & Nelson, 2014), being a highly scale and context-dependent (Siders, 2019). Furthermore, its perceptions are also subjective (Ziervogel, Cowen, & Ziniades, 2016), meaning that adaptive capacity should be considered not as an abstract and metaphorical criterium, but as situated within a particular social and cultural context (Cote & Nightingale, 2011).

On first glance, there is a clear difference between the US and EU perspectives: while the former is rooted in engineering, the latter is in sustainability. We noticed also an interesting mismatch when it comes to the ‘bouncing back’ and ‘forward’ approaches. When asked straightforwardly, more people chose ‘bouncing forward’ rather than ‘bouncing back’ (Fig. 6A). However, when we scrutinized the characteristics and strategies listed through respondents answers (to open-ended questions), it curiously turned out that answers associated with bouncing back (e.g. withstand, resist, counter, absorb) were far more mentioned than those related to bouncing forward (e.g. derivatives of the words transition, transformation, change, improve) (Fig. 6B). This suggests that while the definition of urban resilience related to the ‘bouncing forward’ metaphors have become more popular, different actors might still have varying ideas about what that entails, thus, unknowingly, implementing urban resilience strategies related to ‘bouncing back.’

From this mismatch various considerations could emerge:

- 1) Lack of consistency among European practitioners regarding the implementation of transformational resilience approaches.

While practitioners have grasped the trend of linking resilience to transformation and change (rather than robustness only), there is an emerging inconsistency about how this new paradigm relates to practice. This deduction comes from the above-mentioned misalignment between the perspectives and characteristics and can be illustrated by the quotation from a Spanish practitioner, “resilience is about bouncing back because cities need to anticipate and transform themselves to be

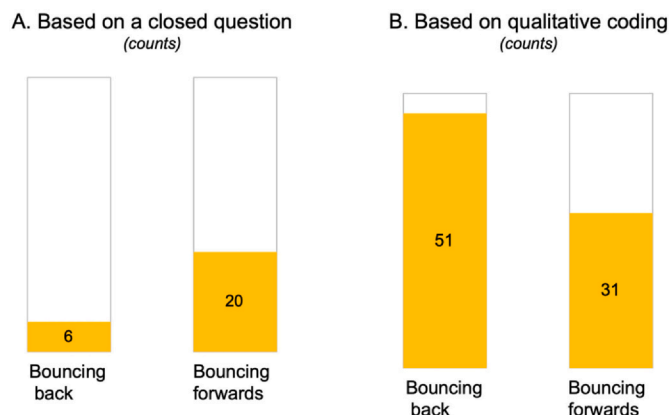


Fig. 6. Comparing the results from closed-ended and open-ended questions.

resilient for eventual shocks.” Indeed, while papers linking resilience and transformations call for systemic transitions toward sustainability (Elmqvist, 2014; Matyas & Pelling, 2015), practitioners' perspectives are far from this framing, relating transformation to improvement, learning and change for better disaster resilience preparation and reaction (see Section 4.2).

- 1) Urban resilience theory fails to properly guide implementation through resilience characteristics and principles. Recent literature has stated that while defining urban resilience is easy, the consensus over a clear set of characteristics is still a challenge (Sharifi et al., 2016, Coaffee et al., 2018). There are limits of applicability of social-ecological resilience principles in urban systems (Chelleri, 2016); while redundancy and connectedness are key characteristics of resilient social-ecological systems, they are not, however, always good indicators for resilience of the sub-urban system (Olazabal, Chelleri, & Sharifi, 2018). Critically looking into the paper results, it seems clear that most of the urban resilience characteristics do not further describe resilience features, thus failing to guide the implementation of different approaches. Instead, most of the characteristics mentioned reflect attributes linking resilience with social justice through participatory planning, inclusive urban development and other measures (see Fig. 5).

A vivid example of this tendency is the lack of weight and strength that the concept of “decentralisation” received throughout our study. Decentralisation has indeed the metaphorical and practical potential for precisely driving the implementation of a specific urban resilience (transformative) approach toward empowerment. It does so by shifting the control over urban infrastructure and resources from a central institution to the local level, by promoting the redistribution of benefits as well as local involvement in planning and decision making processes (Sharifi et al., 2017). However, it was barely mentioned among the characteristics of a resilient city in our questionnaire responses and interviews. This confirms our main deduction that while urban resilience characteristics are still not mature to consistently drive resilience implementation, they are instrumental in establishing bridges among disciplines or goals.

- 1) “Bouncing forward” (or transformational resilience) most of the time is not aligned with sustainability. As introduced through the literature review (Section 2.2), urban transformations are not only related to sustainability (Wolfram & Frantzeskaki, 2016) but also to theoretical discourses around evolutionary (or dynamic and transformational) resilience (Chelleri & Olazabal 2012, Davoudi et al., 2013, Elmqvist, 2014, Pelling, O'Brien, & Matyas, 2014). Our results confirmed the perspective raised by Wolfram and Frantzeskaki (2016) about transformational aspects are not always sustainable. Indeed, when asked about how to bounce forward, practitioners gave answers mostly related to enhancing preparedness to future shocks. Notwithstanding some recent papers addressing and problematizing the linkages between urban resilience and sustainability (Romero-Lankao, Gnatz, Wilhelmi, & Hayden, 2016; Zhang & Li, 2018), our survey results highlighted two different perspectives among the respondents. On the one hand, resilience and sustainability have different goals but do not necessarily contradict each other (Section 4.1.3). On the other hand a lack of incentives and political agendas were listed among the barriers for aligning urban resilience to a bouncing forward implementation. This is clear evidence highlighting the gap between what aligns resilience and sustainability in theory versus what it implies in practical terms. Our results suggest that while international policy agendas use urban resilience and sustainability almost interchangeably (Elmqvist et al., 2019), on-the-ground perceptions are far from sustainability transformations or transitions, as even references to transformative resilience approaches boil down to bouncing back and business

continuity.

5.2. Emerging questions for urban resilience research

The overall result of this paper identifies that across Europe the understanding of urban resilience has been evolving toward the integration of different perspectives. However, when it comes to the details about which are the criteria and principles driving implementation, both academics and practitioners provide different (and sometimes confusing) responses. Some critical literature suggests the very idea of resilience should be abandoned as it relies on technocratic fixes, economic growth and green consumption (Bonds, 2018; Kaika, 2017; MacKinnon & Derickson, 2013; Neocleous, 2013). Throughout our interviews, however, resilience was perceived as having more social and process-oriented flavours than sustainability, which in turn sounded more technocratic. In light of these results, we suggest a few research lines that address the above-mentioned gaps within the definitions and principles of urban resilience in Europe and beyond:

- 1) Assessing the policy coherence of urban resilience. There is a need to further explore regional and local policy discourses, in particular their evolution and linkages with specific implementation and practice, in order to test the consistency of urban resilience definitions, approaches and implementation. This would unveil the uses and abuses of the concept of resilience, and how much it legitimizes the practices far from the latest understanding of its meaning.
- 2) Deepening the understanding of the relationships between urban resilience and sustainability. Under what conditions do urban resilience and sustainability have the same goal? When do they contradict each other? To what extent does this contradiction could be overcome with help of technological solutions, and when is it a matter of political, ethical and cultural behaviour? Before linking resilience to sustainability in a normative way, we need to understand which specific resilience needs to be decreased in order to allow urban sustainability transformation, and get clearer insights about to whom and which resilience is desirable and why, and which are the costs embedded within these changes? Are we ready, and socially prepared to assume the costs of resilience for transformation and sustainability? The recent COVID-19 pandemic has illuminated the trade-offs between resilience and sustainability and brought to the forefront the questions about which economic sectors are to be supported ((hopefully, innovative green and socially just enterprises and business models) and which not (hopefully, economies and sectors far from helping us to reach a more sustainable future, although supporting thousands of employees).
- 3) Defining and agreeing on the urban resilience principles and characteristics. As described above, social-ecological system resilience principles cannot serve as guiding principles for urban resilience. This deduction necessitates further refinement, exploration and agreement among scholars from different fields and practitioners (which nowadays usually rely on the characteristics and indicators of the Rockefeller 100 Resilient City Framework). Urban resilience implementation is still guided by principles linking resilience to other goals, rather than to its consistent implementation (Section 5.1). Which characteristics and principles should guarantee implementations consistent with the given definitions?

Finally, as proposed recently by Harris et al. (2017), *the process of decision-making should become itself the target for research, innovation and transformation.* Having a rich set of case studies examining decision-making process about urban resilience mainstreaming, which actions are put in place, how these relate to theories, and how theory could guide urban agenda and implementation are the key steps to follow in the future of urban resilience research.

Declaration of competing interest

None.

Acknowledgements

The authors would like to thank the participants of the International Conference “Reframing Urban Resilience Implementation: aligning

Sustainability and Resilience” (Barcelona 2018) and other scholars for taking part in the survey. A special thanks goes to the city practitioners for their availability, to Sara Meerow for providing us her feedback on our research design in order to make it worth comparing with her results from the US, to Mich Zhi Xin for her help in conducting the interviews with practitioners at the urban Resilience European Day 2019 conference and to Brien Shelton for his proofreading.

Annex A: questionnaire

I. Identification of the participant

This questionnaire is part of a study which explores how urban resilience is framed by researchers and practitioners in different fields. The questions in the first section will help us to understand how resilience is related to your work. In the second section you will be asked to share your opinion about characteristics of and strategies for resilience. It will take you about 15 min to complete the survey. We greatly appreciate your participation and express enormous gratitude for your collaboration.

The results of the survey will be processed anonymously. Should you have any questions about the study, please get in touch: baravikova.a@gmail.com

Q1. What is your current professional profile?

- Academic - student/PhD candidate
- Academic - early stage researcher
- Academic - senior
- NGO
- Public-governmental sector
- Industry/ business
- Grassroots / community activist
- Other (please specify)

Q2. Please select up to three areas that your work fits into best.

- Architecture / design.
- Climate change adaptation and mitigation.
- Community work.
- Social sciences.
- Disasters/natural hazards.
- Environmental protection / natural resources management.
- Engineering.
- Urban and regional planning.
- Policy and Governance.
- Smart cities and ICT.
- Sustainable development.
- Other (please specify).

Q3. What is the country of your current affiliation?

Q4. If there are other countries that have been significantly related to your work over the last 5 years, please indicate.

Q5. How central is resilience to your work?

- It is a primary focus of my activities (please specify)
- A great deal of my work is related to resilience (please specify)
- Some part of my work is related to resilience (please specify)
- My work has nothing to do with resilience

Q6. For how long have you been working on resilience?

- Less than a year
- From 1 to 5 years
- from 6 to 10 years
- more than 10 years

Q7. Is the word ‘resilience’ explicitly mentioned in your work?

- Yes
- No (please specify what are the terms used instead of it?)

II. Questions about resilience perspectives Q8 Resilience is primarily about:

- The ability to bounce back
- The ability to bounce forward
- Both bouncing back and bouncing forwards

(Display this Question: If answer is “Both bouncing back and bouncing forwards”)

Q8. What are the key challenges to combine ‘bouncing back’ and ‘bouncing forward’ visions of resilience? (select up to three)?

- Political agenda
- Long-term contracts with utilities
- Business lobbies
- Laws and other specific regulations
- High interconnectedness and scale dependency (it is difficult to innovate alone)
- Lack of financial and/or political incentives to adopt transformative pathways
- Lack of clear indicators of innovation (it is difficult to draw a boundary between bouncing back and forward)
- Social unrest
- Recovery is given highest priority
- There are no substantial challenges
- Other (please specify)

Q9. Could you please list up to 10 characteristics of a resilient city?

Q10. Could you please list up to five most important strategies to achieve city resilience?

Q11. Do you agree or disagree with the following statements on relations between resilience and other concepts?

	Yes	No	Not always
Resilience and sustainability have the same goal			
Resilience is a prerequisite for achieving sustainability			
Resilience and sustainability have distinct different goals and objectives but do not contradict each other			
Resilience is more appealing to policy-makers than sustainability			
Resilience is better defined and more measurable than sustainability			
The concept of smart cities is more relevant for resilience than for sustainability			
All of these are unhelpful buzzwords			

The answers about characteristics and strategies were compiled together, as in many cases they were quite similar or overlapping (when asked about strategies respondents named characteristics — and the other way around.) Even though word frequency does not necessarily suggest significance, it has been automatically calculated because it may allow for the exploration of “any emergent but as yet undetected patterns” (Saldaña, 2009). Totals include stemmed versions of the words (e.g. both ‘sustainability’ and ‘sustainable’). In addition, a number of other, more focused word frequency queries were run. For instance, to examine the first tension (‘bouncing back / forward’), the query contained synonymous and associated words, because terms such as ‘bounce back’ or ‘bounce forward’ are not used that much.

Then, during an iterative reading of the data, thematic codes were identified and elaborated using a combination of inductive and deductive coding (some preliminary categories were established with the help of a literature review; in particular, the list of potential characteristics comes from Meerow et al. (2016) paper). The same unit of text could be included in more than one category. As a way to explore data, some codes and themes (when appropriate) were assigned to the five dimensions for assessments of urban resilience suggested by (Sharifi & Yamagata, 2016): materials and environmental resources, society and well-being, economy, built environment and infrastructure, and governance and institution. These categories, however, were used as an analytical exercise to get a better understanding of the popularity of the topics. They do not directly correspond to the research questions and therefore will not be discussed in this paper.

Annex B List of the interviewees

	City	Country	Affiliation
1	Bologna	Italy	Sustainable development office
2	Glasgow	UK	Resilience office
3	Antwerp	Belgium	Climate adaptation office
4	Copenhagen	Denmark	Resilience office
5	Cascais	Portugal	Innovation department
6	Växjö	Sweden	Strategic planning office
7	Santiago de Compostela	Spain	City council
8	Germany	Germany	Consultancy

9	Helsinki	Finland	climate and environmental office
10	Thessaloniki	Greece	Resilience office
11	Wroclaw	Poland	Sustainable development office
12	Bratislava	Slovakia	Architecture office
13	Barcelona	Spain	Resilience office
14	Bologna	Italy	Innovation office
15	Rome	Italy	Innovation office
16	Rotterdam	Netherlands	Resilience office
17	Dordrecht	Netherlands	City council
18	Glasgow	UK	Spatial planning office
19	Ljubljana	Slovenia	City council
20	London	UK	Private company
21	Bratislava	Slovakia	Ministry of Environment
22	Bologna	Italy	City council
23	Cascais	Portugal	City council
24	Stockholm	Sweden	City council

References

- Achour, N. (2015). Integration of resilience and sustainability: From theory to application. *International Journal of Disaster Resilience in the Built Environment*, 6(3), 347–362.
- Alexander, D. E. (2013). Resilience and disaster risk reduction: An etymological journey. *Natural Hazards and Earth System Sciences*, 13(11), 2707–2716.
- Anderies, J. M., Folke, C., Walker, B., & Ostrom, E. (2013). Aligning key concepts for global change policy: Robustness, resilience, and sustainability. *Ecology and Society*, 18(2).
- Angelovski, I., Shi, L., Chu, E., Gallagher, D., Goh, K., Lamb, Z., ... Teicher, H. (2016). Equity impacts of urban land use planning for climate adaptation: Critical perspectives from the global north and south. *Journal of Planning Education and Research*, 36(3), 333–348.
- Asprone, D., & Manfredi, G. (2015). Linking disaster resilience and urban sustainability: A global approach for future cities. *Disasters*, 39(s1), s96–s111.
- Berkes, F., J. Colding and C. Folke (2003). Navigating Social-ecological Systems: Building Resilience for Complexity and Change.
- Bonds, A. (2018). Refusing resilience: The racialization of risk and resilience. *Urban Geography*. <https://doi.org/10.1080/02723638.2018.1462968>.
- Borie, M., Pelling, M., Ziervogel, G., Hyams, K., & Town, C. (2019). Mapping narratives of urban resilience in the global south. *Global Environmental Change*, 54(December 2018), 203–213. <https://doi.org/10.1016/j.gloenvcha.2019.01.001>.
- Boschma, R. (2014). Towards an evolutionary perspective on regional resilience. *Regional Studies*, 49(5), 733–751.
- Brand, F. S., & Jax, K. (2007). Focusing the meaning(s) of resilience: Resilience as a descriptive concept and a boundary object. *Ecology and Society*, 12(1).
- Brown, K. (2014). Global environmental change I: A social turn for resilience? *Progress in Human Geography*, 38(1), 107–117. <https://doi.org/10.1177/0309132513498837>.
- Carpenter, S., Walker, B., Anderies, J. M., & Abel, N. (2001). From metaphor to measurement: Resilience of what to what? *Ecosystems*, 4(8), 765–781.
- Checker, M. (2011). Wiped out by the “Greenwave”: Environmental gentrification and the paradoxical politics of urban sustainability. *City & Society*, 23(2), 210–229.
- Chelleri, L. (2016). *The urban resilience fallacy: Gaps between theory and practice*.
- Chelleri, L. and M. Olazabal (2012). Multidisciplinary perspectives on Urban Resilience. Bilbao, Basque Centre for Climate Change (BC3).
- Chelleri, L., Schuetz, T., & Salvati, L. (2015). Integrating resilience with urban sustainability in neglected neighborhoods: Challenges and opportunities of transitioning to decentralized water management in Mexico City. *Habitat International*, 48, 122–130. <https://doi.org/10.1016/j.habitatint.2015.03.016>.
- Chelleri, L., Waters, J. J., Olazabal, M., & Minucci, G. (2015). Resilience trade-offs: Addressing multiple scales and temporal aspects of urban resilience. *Environment and Urbanization*, 27(1), 181–198.
- Coaffee, J. (2013). Towards next-generation urban resilience in planning practice: From securitization to integrated place making. *Planning Practice and Research*, 28(3), 323–339.
- Coaffee, J., Therrien, M.-C., Chelleri, L., Henstra, D., Aldrich, D. P., Mitchell, C. L., ... Rigaud, É. (2018). Urban resilience implementation: A policy challenge and research agenda for the 21st century. *Journal of Contingencies & Crisis Management*, 26(3), 403–410. <https://doi.org/10.1111/1468-5973.12233>.
- Cote, M., & Nightingale, A. J. (2011). Resilience thinking meets social theory: Situating social change in socio-ecological systems (SES) research. *Progress in Human Geography*, 1–15.
- Cutter, S. L. (2013). Building disaster resilience: Steps toward sustainability. *Challenges in Sustainability*, 1(2), 72.
- Cutter, S. L., Barnes, L., Berry, M., Burton, C., Evans, E., Tate, E., & Webb, J. (2008). A place-based model for understanding community resilience to natural disasters. *Global Environmental Change*, 18(4), 598–606.
- Davoudi, S., Brooks, E., & Mehmood, A. (2013). Evolutionary resilience and strategies for climate adaptation. *Planning Practice and Research*, 28(3), 307–322.
- Derissen, S., Quaas, M. F., & Baumgartner, S. (2011). The relationship between resilience and sustainability of ecological-economic systems. *Ecological Economics*, 70(6), 1121–1128.
- Dodman, D., Diep, L., & Colenbrander, S. (2017). Making the case for the nexus between resilience and resource efficiency at the city scale. *International Journal of Urban Sustainable Development*, 9(2), 97–106. <https://doi.org/10.1080/19463138.2017.1345740>.
- Doorn, N., Gardoni, P., & Murphy, C. (2018). A multidisciplinary definition and evaluation of resilience: The role of social justice in defining resilience. *Sustainable and Resilient Infrastructure*, 9689, 1–12. <https://doi.org/10.1080/23789689.2018.1428162>.
- Eakin, H., Bojórquez-Tapia, L. A., Janssen, M. A., Georgescu, M., Manuel-Navarrete, D., Vivoni, E. R., & Lerner, A. M. (2017). Opinion: Urban resilience efforts must consider social and political forces. *Proceedings of the National Academy of Sciences*, 114(2), 186–189. <https://doi.org/10.1073/pnas.1620081114>.
- Eakin, H. C., Lemos, M. C., & Nelson, D. R. (2014). Differentiating capacities as a means to sustainable climate change adaptation. *Global Environmental Change*, 27(1), 1–8. <https://doi.org/10.1016/j.gloenvcha.2014.04.013>.
- Elmqvist, T. (2014). Urban resilience thinking. *Solutions*, 5(5), 26–30.
- Elmqvist, T. (2017). Development: Sustainability and resilience differ. *Nature*, 546(7658), 352.
- Elmqvist, T., Andersson, E., Frantzeskaki, N., McPhearson, T., Olsson, P., Gaffney, O., Takeuchi, K., & Folke, C. (2019). Sustainability and resilience for transformation in the urban century. *Nature Sustainability*, 2(4), 267–273.
- Folke, C. (2006). Resilience: The emergence of a perspective for social, ecological systems analyses. *Global Environmental Change*, 16(3), 253–267.
- Folke, C., Carpenter, S. R., Walker, B., Scheffer, M., Chapin, T., & Rockström, J. (2010). Resilience thinking: Integrating resilience, adaptability and transformability. *Ecology and Society*, 15(4), 20.
- Friend, R. and M. Moench (2013). What is the purpose of urban climate resilience? Implications for addressing poverty and vulnerability. *Urban Climate* 6(0): 98–113.
- Gillard, R. (2016). *Questioning the diffusion of resilience discourses in pursuit of transformational change*. (Global environmental politics).
- Godschalk, D. R. (2003). Urban hazard mitigation: Creating resilient cities. *Natural Hazards Review*, 4(3), 136–143.
- Grafakos, S., Trigg, K., Landauer, M., Chelleri, L., & Dhakal, S. (2019). Analytical framework to evaluate the level of integration of climate adaptation and mitigation in cities. *Climatic Change*, 154(1), 87–106.
- Harris, L. M., Chu, E. K., & Ziervogel, G. (2017). Negotiated resilience. *Resilience*, 3293, 1–19.
- Hinkel, J. (2011). “Indicators of vulnerability and adaptive capacity”: Towards a clarification of the science-policy interface. *Global Environmental Change*, 21(1), 198–208. <https://doi.org/10.1016/j.gloenvcha.2010.08.002>.
- Holling, C. (1996). Engineering resilience versus ecological resilience. *Foundations of Ecological Resilience*, 51–66.
- Jha, A. K., Miner, T. W., & Stanton-Geddes, Z. (2013). *Building urban resilience: Principles, tools, and practice*. The World Bank.
- Joseph, J. (2013). Resilience as embedded neoliberalism: A governmentality approach. *Resilience*, 1(1), 38–52.
- Kaika, M. (2017). ‘Don’t call me resilient again!’: The new urban agenda as immunology ... or ... what happens when communities refuse to be vaccinated with ‘smart cities’ and indicators. *Environment & Urbanization*, 1–14. <https://doi.org/10.1177/0956247816684763>.
- Khasalamwa, S. (2009). Is ‘build back better’ a response to vulnerability? Analysis of the post-tsunami humanitarian interventions in Sri Lanka. *Norsk Geografisk Tidsskrift - Norwegian Journal of Geography*, 63(1), 73–88.
- Landauer, M., Juhola, S., & Söderholm, M. (2015). Inter-relationships between adaptation and mitigation: A systematic literature review. *Climatic Change*, 131(4), 505–517.
- Leichenko, R. (2011). Climate change and urban resilience. *Current Opinion in Environmental Sustainability*, 3(3), 164–168.
- Lhomme, S., Serre, D., Diab, Y., & Laganier, R. (2013). Analyzing resilience of urban networks: A preliminary step towards more flood resilient cities. *Natural Hazards and Earth System Sciences*, 13(2), 221–230.
- Liao, K.-H., Le, T. A., & Van Nguyen, K. (2016). Urban design principles for flood resilience: Learning from the ecological wisdom of living with floods in the Vietnamese Mekong Delta. *Landscape and Urban Planning*, 155, 69–78.
- MacKinnon, D., & Derickson, K. D. (2013). From resilience to resourcefulness: A critique of resilience policy and activism. *Progress in Human Geography*, 37(2), 253–270.
- Mannakkara, S., & Wilkinson, S. (2013). Build Back better principles for post-disaster structural improvements. *Structural Survey*, 31(4), 314–327.

- Manyena, B., O'Brien, G., O'Keefe, P., & Rose, J. (2011). Disaster resilience: A bounce back or bounce forward ability? *Local Environment: The International Journal of Justice and Sustainability*, 16(5), 417–424.
- Matin, N., Forrester, J., & Ensor, J. (2018). What is equitable resilience? *World Development*, 109, 197–205. <https://doi.org/10.1016/j.worlddev.2018.04.020>.
- Matyas, D., & Pelling, M. (2015). Positioning resilience for 2015: The role of resistance, incremental adjustment and transformation in disaster risk management policy. *Disasters*, 39(s1), s1–s18.
- McEvoy, D., Lindley, S., & Handley, J. (2006). *Adaptation and mitigation in urban areas: Synergies and conflicts*. Thomas Telford Ltd: Proceedings of the Institution of Civil Engineers-Municipal Engineer.
- Meerow, S., & Newell, J. P. (2016). Urban resilience for whom, what, when, where, and why? *Urban Geography*, 1–21.
- Meerow, S., Newell, J. P., & Stults, M. (2016). Defining urban resilience: A review. *Landscape and Urban Planning*, 147, 38–49.
- Meerow, S., & Stults, M. (2016). Comparing conceptualizations of urban climate resilience in theory and practice. *Sustainability*, 8(7), 701.
- Mulligan, M., W. Steele, L. Rickards and H. Fünfgeld (2016). Keywords in planning: what do we mean by 'community resilience'? *International Planning Studies*: 1–14.
- Neocleous, M. (2013). Resisting resilience. *Radical Philosophy*, 178(February), 2–7. <https://doi.org/10.1080/10643389.2012.728825>.
- Neuman, M. (2005). The Compact City fallacy. *Journal of Planning Education and Research*, 25(1), 11–26.
- Newman, P., Beatley, T., & Boyer, H. (2009). *Resilient cities: Responding to peak oil and climate change*. Island Press.
- Olazabal, M., Chelleri, L., & Sharifi, A. (2018). Is connectivity a desirable property in urban resilience assessments? *Resilience-oriented Urban Planning*. Springer 195–211.
- Olsson, L., Jerneck, A., Thoren, H., Persson, J., & O'Byrne, D. (2015). Why resilience is unappealing to social science: Theoretical and empirical investigations of the scientific use of resilience. *Science Advances*, 1(4), 1–12. <https://doi.org/10.1126/sciadv.1400217>.
- Pearson, L. J., & Pearson, C. J. (2012). Societal collapse or transformation, and resilience. *Proceedings of the National Academy of Sciences*, 109(30), E2030–E2031.
- Pelling, M. (2011). *Adaptation to climate change: From resilience to transformation*. (Taylor & Francis US).
- Pelling, M., O'Brien, K., & Matyas, D. (2014). Adaptation and transformation. *Climatic Change*, 1–15.
- Pizzo, B. (2015). Problematizing resilience: Implications for planning theory and practice. *Cities* 43(0): 133–140.
- Redman, C. L. (2014). Should sustainability and resilience be combined or remain distinct pursuits? *Ecology and Society*, 19(2).
- Rodina, L., Baker, L. A., Galvin, M., Goldin, J., Harris, L. M., Manungufala, T., ... Ziervogel (2017). Water, equity and resilience in southern Africa: Future directions for research and practice. *Current Opinion in Environmental Sustainability*, 26–27(2), 143–151. <https://doi.org/10.1016/j.cosust.2017.09.001>.
- Rodriguez-Nikl, T. (2015). Linking disaster resilience and sustainability. *Civil Engineering and Environmental Systems*, 32(1–2), 157–169.
- Romero-Lankao, P., Gnat, D. M., Wilhelmi, O., & Hayden, M. (2016). Urban sustainability and resilience: From theory to practice. *Sustainability (Switzerland)*, 8(12), 1–19. <https://doi.org/10.3390/su81>.
- Saldaña, J. (2009). *The coding manual for qualitative researchers*. Sage.
- Satterthwaite, D. (2013). The political underpinnings of cities' accumulated resilience to climate change. *Environment and Urbanization*, 25(2), 381–391.
- Satterthwaite, D., & Dodman, D. (2013). Towards resilience and transformation for cities within a finite planet. *Environment and Urbanization*, 25(2), 291–298.
- Schipper, E. L. F. and L. Langston (2015). A Comparative Overview of Resilience Measurement Frameworks.
- Sharifi, A. (2016). A critical review of selected tools for assessing community resilience. *Ecological Indicators*, 69, 629–647.
- Sharifi, A., Chelleri, L., Fox-Lent, C., Grafakos, S., Pathak, M., Olazabal, M., Moloney, S., Yumagulova, L., & Yamagata, Y. (2017). Conceptualizing dimensions and characteristics of urban resilience: Insights from a co-design process. *Sustainability*, 9(6), 1032.
- Sharifi, A., & Yamagata, Y. (2016). Principles and criteria for assessing urban energy resilience: A literature review. *Renewable and Sustainable Energy Reviews*, 60, 1654–1677.
- Siders, A. R. (2019). Adaptive capacity to climate change: A synthesis of concepts, methods, and findings in a fragmented field. *Wiley Interdisciplinary Reviews: Climate Change*, 10(3), 1–18. <https://doi.org/10.1002/wcc.573>.
- Simmie, J., & Martin, R. (2010). The economic resilience of regions: Towards an evolutionary approach. *Cambridge Journal of Regions, Economy and Society*, 3(1), 27–43.
- Smith, A., & Stirling, A. (2010). The politics of social-ecological resilience and sustainable socio-technical transitions. *Ecology and Society*, 15(1), 11.
- Tanner, T., A. Bahadur and M. Moench (2017). Challenges for resilience policy and practice. London.
- The Rockefeller Foundation, A (2014). City Resilience Framework. <http://www.rockefellerfoundation.org/report/city-resilience-framework/> Accessed August 25, 2019.
- Thornton, T., & Combetti, C. (2013). Synergies and trade-offs between adaptation, mitigation and development. *Climatic Change*, 1–14.
- Vale, L. J. (2013). The politics of resilient cities: Whose resilience and whose city? *Building Research and Information*, 42(2), 191–201.
- Vale, L. J. and T. J. Campanella (2005). The resilient city: How modern cities recover from disaster.
- Walker, B. and D. Salt (2006). Resilience thinking: Sustaining ecosystems and people in a changing world.
- Walker, B. and D. Salt (2012). Resilience thinking: Sustaining ecosystems and people in a changing world.
- Walsh-Dilley, M., & Wolford, W. (2015). (Un)defining resilience: Subjective understandings of “resilience” from the field. *Resilience*, 3(3), 173–182. <https://doi.org/10.1080/21693293.2015.1072310>.
- Weichselgartner, J., & Kelman, I. (2015). Geographies of resilience: Challenges and opportunities of a descriptive concept. *Progress in Human Geography*, 39(3), 249–267.
- Wenger, C. (2017). The oak or the reed: How resilience theories are translated into disaster management policies. *Ecology and Society*, 22(3).
- Wolfram, M., & Frantzeskaki, N. (2016). Cities and systemic change for sustainability: Prevailing epistemologies and an emerging research agenda. *Sustainability*, 8(2), 144.
- Zhang, X., & Li, H. (2018). Urban resilience and urban sustainability: What we know and what do not know? *Cities*, 72, 141–148.
- Ziervogel, G., Cowen, A., & Ziniades, J. (2016). Moving from adaptive to transformative capacity: Building foundations for inclusive, thriving, and regenerative urban settlements. *Sustainability*, 8(9), <https://doi.org/10.3390/su8090955>.